COURSE SYLLABUS

Full Stack Development

47 modules | 243 sessions | live classes, recordings, assignments and placement support

01. Orientation, tooling and study plan (Week 1)

- How the program works: live classes, recordings, assignments and doubt sessions
- Installing VS Code, Chrome, Node.js and Git on Windows and macOS
- Setting up your project folder and the terminal basics you will use daily
- How to follow along, take notes and ask useful questions
- Weekly rhythm: class, practice, assignment, review

02. How the web actually works (Week 1)

- Clients, servers and the request and response cycle
- HTTP methods, status codes, headers and cookies
- DNS, domains, hosting and what happens when you type a URL
- Inspecting real requests in the browser Network tab
- Static sites vs dynamic apps vs APIs

03. HTML5 fundamentals (Week 2)

- Document structure, doctype and the head section
- Headings, paragraphs, lists, links and images
- Semantic tags: header, nav, main, section, article, footer
- Tables, media and embedding third party content
- Writing clean, indented, readable markup
- Lab: build a multi-page personal profile site

04. Forms, validation and accessibility (Week 2)

- Input types, labels, placeholders and helper text
- Native validation attributes and custom messages
- Fieldsets, legends and accessible form grouping
- ARIA basics, focus order and keyboard navigation
- Colour contrast, alt text and screen reader habits
- Lab: accessible registration form

05. CSS fundamentals (Week 3)

- Selectors, specificity, inheritance and the cascade
- The box model: margin, border, padding and box-sizing
- Colour, typography, units and custom properties
- Pseudo classes, pseudo elements and transitions
- Debugging styles with DevTools

06. Layout with flexbox and grid (Week 3)

- Flex containers, axes, wrapping and alignment
- Grid templates, areas, gaps and auto-placement
- Sticky headers, sidebars and card grids
- Positioning: relative, absolute, fixed and sticky
- Lab: dashboard layout built twice, with flex and with grid

07. Responsive and modern CSS (Week 4)

- Mobile first thinking and breakpoint strategy
- Fluid type, clamp() and container queries
- Images, aspect ratios and responsive media
- Dark mode with custom properties and prefers-color-scheme
- Animations, transforms and reduced-motion support

- Lab: marketing landing page that works on every screen

08. Git, GitHub and team workflow (Week 4)

- Repositories, commits, staging and history
- Branches, merges and resolving conflicts
- Pull requests, code review and commit message habits
- Forking, cloning, remotes and .gitignore
- Deploying a static site from GitHub

09. JavaScript fundamentals (Week 5)

- Variables, scope, hoisting and strict mode
- Data types, coercion, equality and truthiness
- Operators, conditionals and loops
- Functions, parameters, return values and closures
- Error handling with try, catch and custom errors

10. Arrays, objects and modern syntax (Week 5)

- Array methods: map, filter, reduce, find, some, every
- Object destructuring, spread and rest
- Template literals, optional chaining and nullish coalescing
- Sorting, grouping and shaping real API data
- Immutability and why it matters for UI state

11. DOM, events and browser APIs (Week 6)

- Selecting and creating elements, updating the DOM
- Event listeners, bubbling, delegation and forms
- localStorage, sessionStorage and cookies
- Timers, debounce and throttle
- Lab: task manager with persistence

12. Async JavaScript and APIs (Week 6)

- Call stack, event loop, microtasks and timers
- Callbacks to promises to async/await
- fetch, JSON, error paths and loading states
- AbortController and race conditions
- Consuming a public REST API end to end

13. Modules, npm and build tooling (Week 7)

- ES modules, import and export
- package.json, dependencies and dev dependencies
- npm scripts and environment variables
- Vite, bundling, hot reload and production builds
- Linting and formatting with ESLint and Prettier

14. TypeScript essentials (Week 7)

- Why types help in larger codebases
- Primitives, arrays, objects, unions and literals
- Interfaces, type aliases and generics
- Narrowing, guards and strict mode
- Typing props, state and API responses

15. React foundations (Week 8)

- Components, JSX and rendering rules
- Props, children and composition
- State with useState and deriving values
- Lists, keys and conditional rendering
- Thinking in components: splitting a page into pieces

16. React state, forms and effects (Week 9)

- Controlled and uncontrolled inputs
- Form validation and error display patterns
- useEffect, dependencies and cleanup
- useReducer for complex state
- Context for shared state

17. React data, routing and performance (Week 10)

- React Router routes, params and nested layouts
- Loading, error and empty states
- Caching with React Query or SWR
- useMemo, useCallback and memo when they actually help
- Code splitting and lazy loading

18. Tailwind CSS and UI systems (Week 10)

- Utility first styling and design tokens
- Responsive and state variants
- Building a reusable button, card and dialog
- Accessible dialogs, dropdowns and focus traps
- Dark theme and design consistency

19. Next.js App Router (Week 11)

- Server and client components
- File based routing, layouts and loading UI
- Data fetching, caching and revalidation
- Route handlers and server actions
- Metadata, SEO and image optimisation

20. Node.js runtime essentials (Week 12)

- Event loop, non blocking IO and streams
- Modules, file system and path utilities
- Environment configuration and process signals
- Reading and writing JSON data safely
- Structuring a Node project for growth

21. Express and REST API design (Week 12)

- Routing, middleware and error middleware
- Controllers, services and validation layers
- Status codes, pagination, filtering and sorting
- Versioning, documentation and Postman collections
- Rate limiting, CORS and request logging

22. MongoDB and Mongoose (Week 13)

- Documents, collections and BSON types
- CRUD with Mongoose models and schemas
- Relationships: embedding vs referencing
- Aggregation pipeline for reports
- Indexes, performance and data validation

23. SQL and PostgreSQL (Week 14)

- Tables, keys, constraints and normalisation
- SELECT, JOIN, GROUP BY and subqueries
- Transactions and ACID behaviour
- Indexes and reading query plans
- Using PostgreSQL from Node with Prisma

24. Authentication and authorisation (Week 15)

- Password hashing with bcrypt and salt rounds
- Sessions, cookies and JWT compared
- Refresh tokens and expiry handling
- Role based access control and route guards
- OAuth basics with Google sign in

25. Security hardening (Week 15)

- OWASP Top 10 walkthrough with examples
- Input validation and injection prevention
- XSS, CSRF and content security policy
- Secrets management and least privilege
- Dependency audits and safe upgrades

26. Testing and quality (Week 16)

- Unit tests with Vitest and assertion design
- Testing React components and hooks
- API integration tests and test databases
- End to end tests with Playwright
- Coverage, flakiness and testable architecture

27. Capstone 1: full stack e-commerce app (Weeks 17-18)

- Requirement gathering and feature slicing
- Data modelling for products, carts and orders
- Building the catalogue, cart and checkout flow
- Admin panel for products and orders
- Payments integration sandbox and order emails
- Deployment, demo and code review

28. Java foundations (Week 19)

- JDK, JVM, IntelliJ setup and first program
- Types, operators, control flow and methods
- Objects, classes, inheritance and interfaces
- Collections, generics and streams
- Exception handling and file IO

29. Spring Boot REST services (Week 20)

- Project structure, starters and configuration
- Controllers, DTOs and validation
- Service layer and exception handling
- Spring Data JPA repositories
- Testing with JUnit and MockMvc

30. JPA, Hibernate and MySQL (Week 20)

- Entities, relationships and fetch strategies
- JPQL and criteria queries
- Transactions and optimistic locking
- Schema migrations with Flyway
- Building a production style CRUD API

31. Spring Security (Week 21)

- Filter chain and authentication managers
- JWT based stateless authentication
- Method level roles and permissions
- CORS, CSRF and password encoders
- Securing a multi role admin API

32. Python foundations (Week 22)

- Syntax, types, control flow and functions
- Lists, dicts, sets and comprehensions
- Modules, packages and virtual environments
- File handling, JSON and CSV
- Object oriented Python and type hints

33. Django and DRF (Week 23)

- Project and app structure, settings and URLs
- Models, migrations and the admin site
- Views, templates and forms
- Django REST Framework serializers and viewsets
- Authentication, permissions and throttling

34. Python for automation and data (Week 23)

- Working with requests and scraping the safe way
- Pandas basics for tabular data
- Cleaning, joining and summarising datasets
- Scheduling scripts and logging
- Packaging a reusable internal tool

35. AI integration: OpenAI and Gemini APIs (Week 24)

- How large language models work at a practical level
- Keys, quotas, cost control and environment setup
- Chat completions, streaming and system prompts
- Structured output with JSON schema
- Embeddings, vector search and retrieval basics
- Guardrails, moderation and fallback behaviour

36. AI assisted development (Week 24)

- Using AI for scaffolding, refactors and tests
- Reviewing generated code instead of trusting it
- Writing good prompts with context and constraints
- Privacy, licensing and when not to use AI
- Keeping a human review checklist

37. Building an AI feature end to end (Week 25)

- Choosing the feature: assistant, summariser or search
- Designing prompts, tools and data flow
- Server side API calls and secret handling
- Streaming responses into the UI
- Evaluating quality and logging failures
- Shipping it behind a feature flag

38. Docker and containers (Week 26)

- Images, containers, layers and registries
- Writing a production Dockerfile for Node
- Compose for app plus database
- Volumes, networks and environment files
- Image size, caching and security basics

39. CI/CD with GitHub Actions (Week 26)

- Workflow files, jobs, steps and runners
- Linting, type checks and tests on every push
- Build artefacts and release tags
- Environment secrets and approvals
- Automatic deploys and rollback plans

40. Cloud deployment and operations (Week 27)

- Frontend hosting on Vercel and Netlify
- API hosting on Render, Railway and AWS basics
- Managed databases, backups and connection pooling
- Domains, HTTPS and DNS records
- Cost awareness and scaling knobs

41. Production debugging and observability (Week 27)

- Reading logs and structured logging
- Health checks, uptime monitors and alerts
- Error tracking with Sentry style tools
- Tracing a slow request across the stack
- Incident notes and postmortems

42. Capstone 2: SaaS dashboard with AI assistant (Weeks 28-29)

- Multi tenant data modelling and auth
- Billing and plan based feature limits
- Team invites, roles and audit trail
- Embedding an AI assistant with citations
- Load testing and performance tuning
- Final demo, documentation and handover

43. System design for interviews (Week 30)

- Requirements, constraints and back of envelope numbers
- Caching, queues and background jobs
- Scaling reads, writes and databases
- Designing a URL shortener and a chat app
- Trade off conversations interviewers listen for

44. Data structures and algorithms (Weeks 30-31)

- Complexity, arrays, strings and two pointers
- Hashing, stacks, queues and linked lists
- Recursion, trees and binary search
- Sorting, heaps and greedy patterns
- Graphs, BFS, DFS and dynamic programming basics
- Pattern based problem practice sets

45. Aptitude and logical reasoning (Week 31)

- Percentages, ratios, averages and profit and loss
- Time, speed, distance and work
- Permutations, combinations and probability
- Number series, coding decoding and puzzles
- Timed mock tests with review

46. Resume, portfolio and LinkedIn (Week 32)

- Writing impact bullets with numbers
- Project descriptions that survive screening
- GitHub cleanup, README files and live links
- LinkedIn profile and referral outreach
- Portfolio site walkthrough

47. Mock interviews and placement drive (Week 32)

- DSA round simulation with feedback
- Machine coding round practice
- System design and project deep dive rounds
- HR round, salary discussion and offer etiquette

- Weekly drives, referrals and follow up tracker

Fees, batch dates and mode (online / classroom): confirm the current numbers on the website or with the admissions team before enrolling. Yantrix Technology, Pune.